

Learning objectives

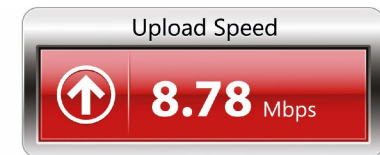
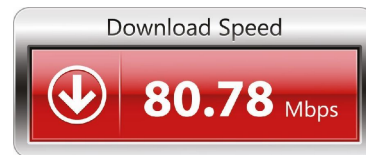
In this lesson you will learn to:

- Define the meanings of the terms 'bandwidth' and 'latency'
- Explain how bandwidth and latency affect the performance of a network
- Use bits per second (bps) to describe network speed
- Construct expressions involving file size, transmission rate and time.

For more information and additional learner activities see Chapter 4.1 of the student book.

Network speed

How fast is your network?



What is an acceptable speed?



What does speed mean?

The term 'download speed' refers to how much data can be downloaded per second over a connection.

There are a number of factors that can affect the speed of a connection. The two most important are the **bandwidth** and **latency**.

Bandwidth and latency

Bandwidth is the maximum amount of data that can be transmitted over a connection per second. It is a measurement of capacity. It is measured in bits per second (**bps**).

Latency is the time taken for a transmission to reach its destination. It is measured in milliseconds.

Gamers may be familiar with the term '**ping**'. This is a way of measuring latency. It times a single transmission as it travels to a server and back again.

Measuring speed

We measure network speed in bits per second (bps).

Modern networks have high bandwidths, measured in millions of bits per second.

The following units are used to measure network speeds.

Units	Abbreviation	bits per second
bits per second	bps	1
kilobits per second	Kbps	1000
megabits per second	Mbps	1 000 000
gigabits per second	Gbps	1 000 000 000

File size recap

Previously you learned how we measure the size of files. Notice that these units are different to the ones we use for speed.

Unit	Abbreviation	Bytes	Equivalent to
bit			1 bit
nibble			4 bits
byte		2^0 bytes	8 bits or 2 nibbles
kibibyte	KiB	2^{10} bytes	1024 bytes
mebibyte	MiB	2^{20} bytes	1024 kibibytes
gibibyte	GiB	2^{30} bytes	1024 mebibytes
tebibyte	TiB	2^{40} bytes	1024 gibibytes

Calculating transmission times

In order to calculate the estimated time for a file to be transmitted, you need to construct an expression.

The basic formula for 'time to transmit a file' is:

$$time = \frac{file\ size\ (in\ bits)}{network\ speed\ (in\ bps)}$$

In order to construct the expression, you need to convert the file size into bits and network speed into bits per second.

Worked example

How long will it take to download a 20MiB file over a 12 Mbps connection?

1. Convert the file size to bits

$$20 \times 8 \times 1024 \times 1024$$

2. Convert the speed to bits per second

$$12 \times 1000 \times 1000$$

3. Arrange the size and speed expressions

$$time = \frac{20 \times 8 \times 1024 \times 1024}{12 \times 1000 \times 1000}$$

Calculating file size and speed

The formula can be rearranged to calculate any of its values:

$$time = \frac{file\ size\ (in\ bits)}{network\ speed\ (in\ bps)}$$

$$size = time\ (in\ seconds) \times speed\ (in\ bps)$$

$$speed = \frac{file\ size\ (in\ bits)}{time\ (in\ seconds)}$$

Wrap up: you have learned how to...

- Define the meanings of the terms 'bandwidth' and 'latency'
- Explain how bandwidth and latency affect the performance of a network
- Use bits per second (bps) to describe network speed
- Construct expressions involving file size, transmission rate and time.